Question No. 61
Differentiate between the guided and unguided transmission media. Give at least 2 examples of each.

1. **Guided** Transmission Media involves the transmission of data through physical paths like cables. Examples:
   - Twisted Pair Cable used in telephone lines and LANs.
   - Fiber Optic Cable: High-speed data transmission with minimal signal loss.
2. **Unguided** Transmission Media involves the transmission of data without physical paths like cables. Often referred to as wireless transmission. Examples:
   - Wireless LANS, Radio, Television, Satellite internet, etc.

Question No. 62
Outline 2 advantages and 2 disadvantages of Fiber Optics.

Advantages:

High Speed: Supports extremely fast data transmission rates.

Long Distance: Minimal signal loss, making it ideal for long-distance communication.

Immune to Electromagnetic Interference (EMI): Resistant to electrical noise.

Disadvantages:

High Cost: Expensive to install and maintain.

Fragility: Glass fibers are more delicate compared to copper wires.

Difficult Installation: Requires specialized skills and equipment for setup and repair.

Question No. 63
Complete the table for the given wireless bands.

Band	Range	Propagation	Application(s)
Very Low Frequency VLF	3 – 30 kHz	Ground	Long-range radio navigation
Middle Frequency MF	300 kHZ – 3 MHz	Sky	AM radio
Very High Frequency VHF	30 – 300 MHz	Sky, Line of Sight	FM radio, VHF Television
Ultra High Frequency UHF	300 MHz – 3 GHz	Line of Sight	UHF Television, Cellular Phone
Superhigh Frequency SF	3 – 30 GHz	Line of Sight	Satellite

Question No. 66
Outline the three processes take place in the Virtual-Circuit.

// A virtual-circuit network is neither a pure circuit-switched network nor a packet-switched network, but instead a virtual circuit network is a mix of circuit-switched and packet-switched networks.

i. **Connection Setup:** a logical path is established between the sender and receiver, and resources may be reserved
ii. **Packetization:** involves dividing data into packets/cells and adding headers (VCIs)
iii. **Routing:** the packets are then forwarded to the destinations along this established path based on the VCI (not dynamically routed like in a datagram packet-switched network)

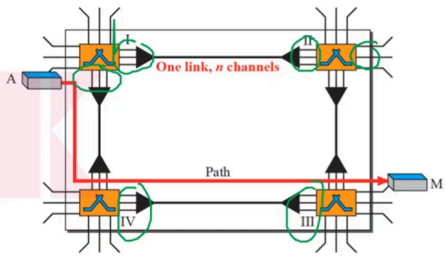
Question No. 65
What are the three phases in a Circuit-Switched Network?

A set of switches connected by physical links where each link is divided into 'n' channel by using FDM or TDM. Following are the three phases in a circuit-switched network.

i. **Connection Establishment:** A dedicated communication path is set up between sender and receiver before data transfer begins.

ii. **Data Transfer:** Data is transmitted over the established path, ensuring a continuous and reliable flow.

iii. **Connection Termination:** The connection is released after the communication is complete, freeing up network resources.



Question No. 64
Differentiate between Circuit and Packet switching.

Connecting multiple devices and forwarding the data frame to the destined device is termed as switching.

Aspect	Circuit Switching	Packet Switching
Connection	A dedicated path is established between sender and receiver.	Data is broken into packets and each packet is sent independently.
Resource Allocation	Reserved for the entire session.	Used as needed, shared among users.
Efficiency	Inefficient; resources remain idle during inactivity.	Highly efficient; resources are used dynamically.
Delay	Minimal delay once the connection is established.	Variable delay due to packet routing.
Example	Telephone networks.	Internet (e.g., email, web browsing).

Question No. 67
What type of communication takes place at the Data Link layer?

The communication at the Data Link layer is **node-to-node**. If a packet is to travel from a host to another host, it needs to pass through various networks from the source to destination.

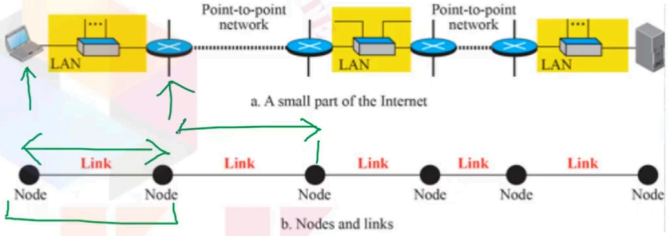


Figure showing Nodes and Links

Question No. 68
How collisions are avoided in the CSMA/CA?

CSMA/CA was invented for Wireless Networks. Collisions are avoided through using of the three strategies:

i. **The Interframe Space:** station waits for a period of time called interframe space IFS. IFS is variable and used to prioritize stations. E.g., station with shorter IFS has higher priority

ii. **The Contention Window:** amount of time station waits to get the medium idle. Each time it finds it busy it increases this time exponentially (doubles), a kind of back off strategy

iii. **Acknowledgements:** The positive acknowledgment and the time-out timer can help guarantee that the receiver has received the frame.

Question No. 69
Name any five services provided by the Data Link layer?

The data link layer received services from physical layer and provide services to network layer. Here are the five services it provides:

i. **Framing:** the process of dividing data into small, manageable units called frames, adding headers and trailers to define start, end, and control information for reliable transmission.

ii. **Encapsulation:** involves wrapping data in a specific format or protocol before transmission.

iii. **Flow Control:** control is the regulation of data transmission between devices.

iv. **Error Control:** implementing techniques to detect and correct errors occur during transmission.

v. **Congestion Control:** involves managing and preventing network congestion to maintain optimal performance.

Network congestion happens when a network is overloaded with traffic.

Question No. 72
What are the three random access protocols?

As the name suggests, no station has any superiority in the network. Any station has equal rights to access the medium and send data. Here are the three random access protocols:

i. **ALOHA:** developed in early 70's, a simple protocol where devices can transmit data whenever they need, but collision may occur and retransmission is required
ii. **CSMA/CD:** stops transmission upon collision, initiates back off, and retries.
iii. **CSMA/CA:** employs strategies like waiting before transmission to minimize collisions.

Question No. 71
Give any three character tics of WLAN that differentiate it with LAN.

Here are the characteristics of Wireless LAN that differentiate it with traditional LAN:

i. **Access method:** WLAN uses the CMSA/CA as while Ethernet LAN uses CSMA/CD
ii. **Mobility:** WLAN allows devices to move freely within coverage area
iii. **No physical** cables/connections needed
iv. **Dynamic Network:** devices have dynamic connection, easily join or leave the network

Question No. 70
The Ethernet is a unreliable technology. Justify your answer.

What are the limitations of the Ethernet?

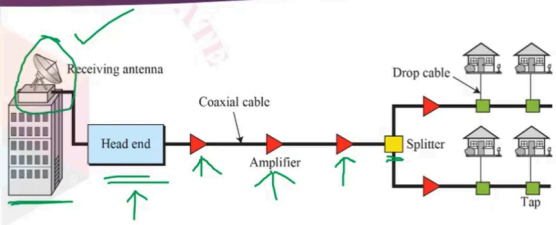
What could be the disadvantages of Ethernet? Criticism? Etc.

Yes, the ethernet is a unreliable technology, considering its characteristics well aligned with the unreliable services as stated below.

i. No connection establishment or teardown process
ii. The sender may overwhelm the receiver with frames, and frames are dropped
iii. If a frame drops, the sender will not know about it unless we are using TCP (Transport)
iv. Ethernet is unreliable like IP and UDP
v. If a frame is corrupted, the receiver silently drops it

Question No. 75
What is the cable networks? How did it evolve?

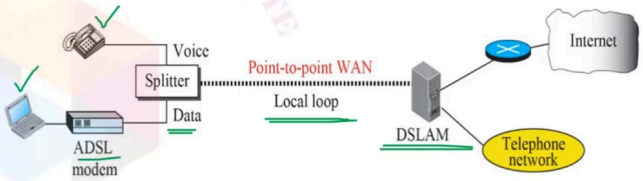
Cable TV network was initially started to provided remote subscribers access to TV. Initially, it was known as **CATV – community antenna television**.



An antenna at the top pf a tall hill or building receiving the signals from tv stations and then distributed to subscribers.

Question No. 74
What is DSL technology? Name its different technologies.

Digital Subscriber DSL was developed by the telephone companies to provide high-speed digital data transmission over the existing telephone lines. DSL technology is a set of technologies:



i. SDSL: Symmetric Digital Subscriber Line
ii. ADSL: Asymmetric Digital Subscriber Line
iii. HDSL: High Bitrate Digital Subscriber Line.
iv. VDSL: Very High Bitrate Digital Subscriber Line

Question No. 73
What are the three controlled access methods in the networks?

Following are the three controlled access methods:

i. **Reservation:** time is divided into internals, and each station needs to make reservation (reservation frame) before sending data (data frame)

ii. **Polling:** works with topologies where one station serves as primary while others are secondary devices. All message exchange must be made through the primary which controls the link while secondary follows the instructions.

iii. **Token Passing:** organized in a logical ring, a station having the token (special packet) has right to send data, here token, priority management, etc. is required

Question No. 77
Which can help guaranteed that the receiver has received the frame?

To guarantee that the receiver has received the frame, an Acknowledgment (ACK) mechanism is required. This is typically implemented at higher layers, as the data link layer itself may not provide such guarantees.

- **Acknowledgments (ACKs):**The receiver sends an acknowledgment to the sender upon successful receipt of a frame. This is often used in protocols like Stop-and-Wait ARQ or Sliding Window Protocol.
- **Retransmission Timer (Time out):** If the sender does not receive an acknowledgment within a specific time, it retransmits the frame.

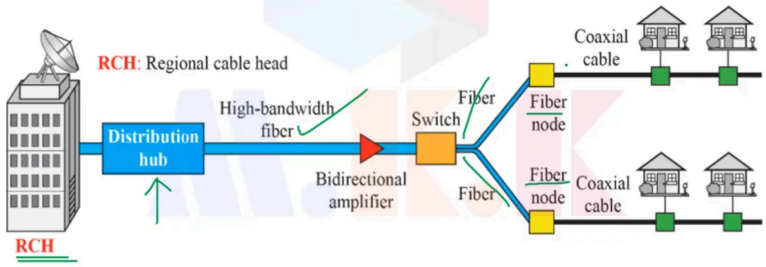
Question No. 76
In the context of error detection, if a codeword is cyclically shifted, what would be the result?

In the context of error detection in the field of data communication, if a codeword is cyclically shifted:

- The result is still a valid codeword in a cyclic code.
- Cyclic codes, such as CRC (Cyclic Redundancy Check) codes, are designed so that any cyclic shift (left or right) of a valid codeword will also produce another valid codeword.

This property is useful in detecting errors during data transmission, as the cyclic nature of the code ensures consistent verification.

Question No. 75
What is the cable networks? How did it evolve?



Question No. 80
Differentiate between a Hub and a Switch.

Aspect	Hub	Switch
Definition	A simple device	A more advanced device
OSI Layer	Operates at the Physical Layer (Layer 1)	Operates at the Data Link Layer (Layer 2)
Data Transmission	Broadcasts data to all connected devices	Uses MAC addresses to forward data only to the specific destination device.
Speed	Typically slower due to shared bandwidth.	Faster due to intelligent data forwarding and collision prevention.
Efficiency	Less efficient as all devices receive unnecessary data.	More efficient as data is sent only to the intended recipient.

Question No. 79
How does block code work? How to calculate the no. of valid codewords?

Here is the simple process that block coding uses:

i. We divide our message into blocks, each of 'k' bits, called data words

ii. We add 'r' redundant bits to each block to make the length 'n = k + r'

iii. The resulting 'n-bit' blocks are called codewords

Let us assume that k=2: Number of data bits (message size). The number of valid codewords in a block code is given by 2^k.

Hence, $2^k = 2^2 = 4$ ∴

Question No. 78
Complete the below table for Ethernet.

Description	Standard	Length	Data Rate
Standard Ethernet	10Base-T	100 meters	10Mbps
Fast Ethernet	100Base-TX	100 meters	100Mbps
Gigabit Ethernet	1000Base-T	100 meters	1 Gbps
10 Gigabit Ethernet	10GBase-T	100 meters	10 Gbps

Question No. 83
What are the four functional layers of SONET?

The SONET standard includes four functional layers:

- The Path Layer
- The Line Layer
- The Section Layer
- The Photonic Layer

OSI Layers	SONET Layers
Data Link	Path
	Line
	Section
Physical	Phonic

Question No. 82
Calculate the transmission delay if a file of 1 Megabit is sent over Ethernet.

The formula to calculate transmission delay is:

$$\text{Transmission Delay} = \frac{\text{Data Size (bits)}}{\text{Transmission Rate (bits per second)}}$$

$$\textbf{Transmission Delay} = \frac{1{,}000{,}000}{10{,}000{,}000} = \textbf{0.1 seconds or 100 ms}$$

Note:

Transmission delay is independent of the physical distance between devices. It only depends on the data size and the transmission rate of the medium.

Question No. 81
What is the Minimum Hamming Distance for these words? d(10101, 11110)

The Hamming distance is the number of positions where the corresponding bits of two binary strings differ.

Here we are given the word as: d(10101, 11110)

We need to compare each bit position as:

1	0	1	0	1
1	1	1	1	0

Let us count the difference, which is 3.

Hence, The Hamming distance d(10101,11110) = 3

Question No. 85
what are Data Link sub layers and what Service do these provide?

The Data Link Layer is divided into two sublayers:

Logical Link Control (LLC)	Media Access Control (MAC)
Error Detection and Correction	Access Control
Flow Control	Addressing
Multiplexing	Frame Transmission
Framing	Collision Handling

Question No. 84

A dataword consists of 3 bits. Using the even parity check coding method, calculate the parity bit for each of the following datawords.

Dataword	Codeword
101	
110	
111	
000	
011	

The even parity check ensures that the total number of 1's in the codeword (dataword + parity bit) is even.

Add the parity bit (0 or 1) to make the total number of 1's even.

Question No. 88
What are the three channelization protocols?

The available bandwidth of a link is shared in time, frequency, or through code, among different stations using channelization protocols. These include:

i. **Frequency** Division Multiple Access (FDMA): the available bandwidth is divided into frequency bands and each station is allocation a band to send data

ii. **Time** Division Multiple Access (TDMA): Stations share the bandwidth of the channel in terms of time and each station is allocated a time slot to send data

iii. **Code** Division Multiple Access (CDMA): allows multiple users to share the same frequency band and time by assigning unique codes to each user, enabling simultaneous transmission.

Question No. 87
Identify the type of access method for the following.

Protocol	Access Method
ALOHA	Random Access
Polling	Controlled Access
Reservation	Controlled Access
CSMA/ CA	Random Access
CSMA/ CD	Random Access

Question No. 86
"The station A send data packet to station B. Station A can send packet again to B when it receive ACK reply from B ". What protocol is this?

The given situation clearly state that it is Stop-and-Wait Protocol ensures reliable data transmission. Station A sends one data packet to Station B and waits for an acknowledgment (ACK) from Station B before sending the next packet.

This process ensures that each packet is successfully received before proceeding with the next one. It is commonly used in reliable communication systems at the Data Link Layer and Transport Layer.

Question No. 91
Find the data rate of sts-1 frame.

The STS-1 (Synchronous Transport Signal level 1) is the basic unit of the SONET (Synchronous Optical Networking) standard.

Data Rate of STS-1: The data rate of an STS-1 frame is **51.84 Mbps** (megabits per second). This rate is achieved by transmitting a frame that consists of 810 bytes of data (or 6480 bits) at a frequency of 8000 frames per second.

Question No. 90
Differentiate between inter-LATA and intra-LATA services.

i. **Intra-LATA** Services: Services offered by Telephone companies inside a LATA are called Intra-LATA services. Usually, Local Exchange Carrier (LEC) handle such services.

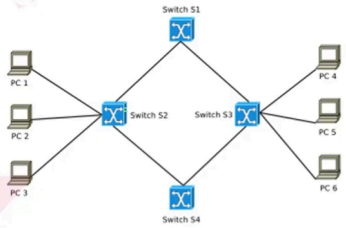
ii. **Inter-LATA** Services: Services offered by Telephone companies between different LATAs are called Inter-LATA services. Interexchange Carriers (IXCs) are the companies that handle intra-LATA services.

LATA is a term used in U.S tekeommunication regulations. A LATA is a geographical area established for the provision and administration pof communication services.

Question No. 89
Consider below network topology and identify the problem it may lead to.

In the given topology, we can see that there are redundant switches configured in the network that is creating a redundant paths. For example, if PC2 wants to send data to PC5, the frame can be forwarded using the path S2→S1→S3 or S2→S4→S3.

Such a situation will lead to a loop in the network. Switches are efficient and intelligent devices which run Spanning Tree Algorithm STP to cater to such problems. STP blocks either path to avoid such loop in the network.



Question No. 94
Names of three channels used in a telephone network by using DSL technique.

In a Digital Subscriber Line (DSL) network, the available frequency spectrum is divided into multiple channels to facilitate simultaneous transmission of voice and data. The primary channels are:

i. **Voice** Channel: This channel operates within the frequency range of 0 to 4 kHz and is used for traditional voice communication.

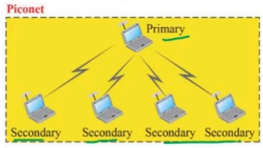
ii. **Upstream** Data Channel: Typically operating between 25.875 kHz and 138 kHz, this channel is used for data transmission from the user to the service provider.

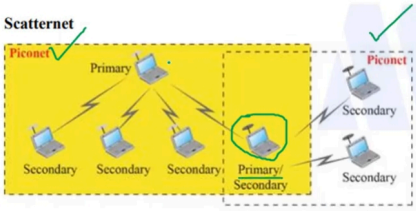
iii. **Downstream** Data Channel: Operating from 138 kHz up to 1.104 MHz, this channel is used for data transmission from the service provider to the user.

Question No. 93
Bluetooth defines two types of networks. What are those?

Bluetooth is a wireless LAN technology designed to connect devices of different functions when they are at a short distance from each other. Its architecture defines two types of network:

Piconet: one device serves as the Primary (Master) station, while up to seven other devices function as Secondary (Slave) stations.





Scatternet: multiple piconets are interconnected, allowing devices to participate in multiple piconets simultaneously. Each piconet consists of one Primary (Master) station and up to seven Secondary (Slave) stations.

Question No. 92
Suppose a coding scheme has a Minimum Hamming Distance (d_{min}) = 5. Calculate how many errors can this code could detect?

A coding scheme with a minimum Hamming distance (d_{min}) of 5 can detect up to 4 errors. This is because a code with a minimum Hamming distance of d can detect up to d - 1 errors.

$d_{min} = 5$

Maximum detectable errors = $d_{min} - 1 = 5 - 1 = 4$ errors ($d_{min} = S + 1$)

Therefore, this code can detect any combination of up to 4 bit errors in a transmitted codeword.

Question No. 97
What services does the Point-to-Point protocol PPP has to offer?

Point-to-point protocol PPP is the most common protocol for point-to-point access. Following are the services PPP does offer:

i. Framing
ii. Authentication
iii. Network Address configuration
iv. Link establishment and data exchange
v. Multi-link PPP address configuration

Question No. 96
Outline the two modes which HDLC provide.

HDLC provides two common transfer modes that can be used in different configurations:

- **Normal Response** Mode (NRM): an unbalanced configuration involving a primary station and one or more secondary stations. Primary initiates the data transfers, while secondary can also send data in the response to primary.
- **Asynchronous Balanced** Mode (ABM): a balanced configuration where each station has equal status where either can initiate the request of data transfer.

Question No. 95
What functions does the Address Resolution Protocol ARP perform?

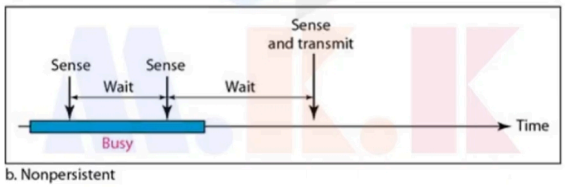
A node has an IP packet to send to another node in a link and it has IP address of the receiving node. However, the IP address of the next node isn't helpful for frame forwarding through a link. Instead, layer-2 (data link) address of the next node is needed.

- Address Resolution Protocol ARP maps an IP address to a layer-2 address. At first, the sender doesn't know the layer-2 address of the receiver, hence, it sends an ARP broadcast message. The destined node responses back with its layer-2 address that is stored inside the local cache of the node (switch or the router) and it is used to forward frames.

Question No. 99
Write down the three persistent methods in the CSMA.

b. Non-Persistent CSMA: (wait – sense – wait)

- If the channel is idle, the station transmits immediately.
- If the channel is busy, the station waits for a random amount of time before checking again.



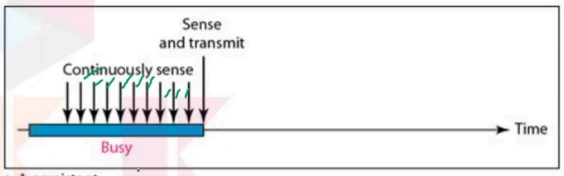
b. Nonpersistent

Question No. 99
Write down the three persistent methods in the CSMA.

CSMA has several persistence strategies to determine how a device should behave when the medium is busy. Here are these:

a. 1-Persistent CSMA: (Continuously sense)

- If the channel is idle, the station transmits immediately.
- If the channel is busy, the station continuously keep checking.



a. 1-persistent

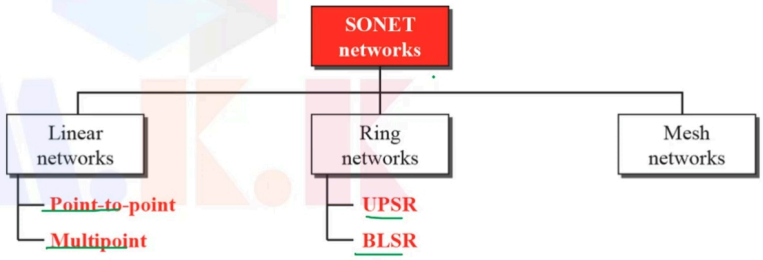
Question No. 98
In the context of Pure ALOHA, what does each of the abbreviation stand for?

Protocol	Access Method
k	Number of attempts
T_p	Maximum Propagation Time
T_{fr}	Average Transmission Time
T_B	Back-off Time: R x T_p or R x T_{fr}
R	Random Number: 0 to $2^K - 1$

Question No. 100
What three categories of networks available in the SONET?

SONET network can be used as a high-speed backbone carrying loads from other networks such as ATM or IP. We can roughly divide SONET networks into three categories:

i. Linear Networks
ii. Ring Networks
iii. Mesh networks



Question No. 100
What three categories of networks available in the SONET?

SONET network can be used as a high-speed backbone carrying loads from other networks such as ATM or IP. We can roughly divide SONET networks into three categories:

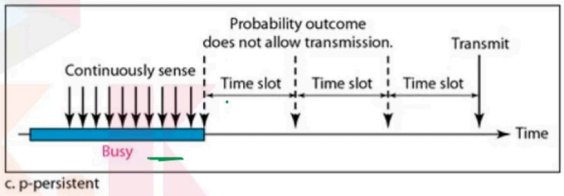
i. Linear Networks
ii. Ring Networks
iii. Mesh networks

Question No. 99
Write down the three persistent methods in the CSMA.

c. P-Persistent CSMA:

- If the channel is idle, the station transmits with probability p.
- If the channel is idle and the station doesn't transmit, it waits for the next time slot (slotted systems) and repeats the process.
- Can transmit if

  random_no. < Probability_Assigned



c. p-persistent